Revision of
Interpretable Multivariate Conformal Prediction with Fast Transductive Standardization

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Yunjie Fan and Matteo Sesia

Response to the Editor and Reviewers

We thank the Editor and both reviewers for their careful feedback. This letter records the changes supported by the integrated revised manuscript. Comments are reproduced in blue italics, followed by a response and completion status. References point to the compiled revised paper. Red author notes identify remaining work and should be resolved before submission.

Summary of Verified Changes

- **Main experiments.** Coordinate-length bar plots, native CQHR, partial heteroskedasticity, and real-data wall-clock times have been added. New 200-split reruns also provide coordinate comparisons on all six real datasets and direct counts of backward searches and global fallbacks. The original two-point energy illustration remains in the body.
- **Supplementary experiments.** Dependence, heterogeneity, target-level, small-sample, heavy-tail, contamination, base-quantile, constant-shift, and shape-template analyses appear in Appendix C.
- **Preservation of earlier evidence.** All historical Gaussian plots, distribution tables, oracle/local-union comparisons, dimension-scaling timings, heavy-tail plots, and the original real-data table are retained in Appendix D. Their original resampling protocol is distinguished from the new fresh-draw experiments.
- **Scope and uncertainty.** The experimental text separates observed coverage from the assumptions of the theorems, retains actual means below target, and reports pointwise Monte Carlo uncertainty without treating one repetition-level standard deviation as a confidence interval for a mean.
- **Supplied non-experimental revisions.** The setup and motivating constructions clarify joint exchangeability, simultaneous versus marginal coverage, vector notation, the parameterized population oracle, and the link-function/local-hypothesis framework. These passages were reviewed but not edited during integration.

Reviewer audit: 11 addressed, 5 partially addressed, and 4 open. The separate `unresolved_reviewer_concerns.md` lists remaining actions and other source-level issues requiring author review.

Response to the Editor

E1. Completeness of the revision and response

Partially addressed

The manuscript was sent to two expert Reviewers, who each noted several positive aspects of the work while also highlighting areas in which it could be improved. One Reviewer felt that the empirical assessment was incomplete and suggested additional results and baselines which could strengthen the work, while the other Reviewer was more positive about the methodology and empirical assessment but expressed some difficulty in digesting elements of the manuscript, suggesting areas for presentational improvement. I would therefore like to invite a resubmission which addresses the two Reviewers' feedback in full, together with a point-by-point response to the Reviewers' reports.

Response. We thank the Editor for the opportunity to revise. The integrated manuscript now includes the requested coordinate-length plots, CQHR and shape-template comparisons, partial heteroskedasticity, contamination stress tests, and main-text runtime evidence. Every previously included experiment is retained: older comparisons are organized in Appendix D, while the original two-point energy example remains in the body.

This letter distinguishes completed work from remaining actions. Of the twenty reviewer points, eleven are addressed, five partly addressed, and four open. The previously missing real-data coordinate and search-branch diagnostics have now been measured in 200 new splits of each dataset. We do not claim that the revision is submission-ready until the non-experimental corrections listed below and in `unresolved_reviewer_concerns.md` have been considered.

E2. Highlighting modified manuscript sections

Partially addressed

For the convenience of the Reviewers, I kindly request that modified sections of the manuscript are indicated using a different colour font.

Response. New and relocated experiment sections are marked in blue. The supplied non-experimental draft also contains blue-marked revisions, but a complete manuscript-wide color audit remains an author check. The original submission was available as a PDF rather than its complete source, and its unchanged-looking passages cannot establish that every modified passage is highlighted. Non-experimental sections were reviewed but not altered during integration.

Response to Reviewer 1

Main Concerns

R1.C1. Intuition for the local construction and rectangular enclosure — Partially addressed

The paper is fairly dense, and the methodological core is harder to parse than it needs to be. The high-level intuition is good, but once the paper gets into the transductive oracle, link functions, worst-case constructions, and the indexing used in the local partition argument, the presentation becomes difficult to follow. I think the paper would benefit from more intuition in Section 3, especially around why the local worst-case construction is the "right" relaxation and how the rectangular enclosure affects efficiency in practice.

Response. We appreciate the request for a clearer conceptual progression. The supplied revision reorganizes the motivating constructions in Section 2 and adopts a common set-valued notation for the population-standardized and transductive procedures. In Section 3.1, the link functions invert a scalar score threshold into coordinate-wise bounds; their monotonicity then permits replacing the unknown oracle threshold by an observable conservative bound. Section 3.4 motivates local working hypotheses and separates partitioning, local bounding, and rectangular enclosure.

The enclosure sacrifices some geometric detail for interpretability and tractability. It is not an unrestricted minimum-volume solution. The original comparison with the unenclosed local union is retained in Appendix D.5, particularly Figure S14; these low-dimensional results are relevant evidence about enclosure tightness. The new coordinate-length plots instead describe informativeness of the final rectangle and do not isolate enclosure loss.

Author completion required. Section 3.4.1 currently infers conditional coverage given a working hypothesis from a marginal oracle guarantee. That does not follow from the stated containment alone. Correct this using the pointwise oracle-acceptance argument, and review the related large-sample tightness language. The exposition has improved, but this point cannot be marked fully resolved while that claim remains.

R1.C2. Assumptions, infinite-variance experiments, and negative residuals — Partially addressed

More discussion of the assumptions and how they relate to the experiments are needed. The theory assumes nonnegative residuals with finite first and second moments, but the appendix also emphasizes robustness under heavy-tailed settings where the population variance is infinite. That may be empirically true, but it creates some tension with the formal assumptions, and the paper should be more explicit about what exactly is proved versus what is only observed experimentally. Along similar lines, the treatment of negative residuals is deferred to workarounds and future work, which is understandable, but it does narrow the immediate scope of the method.

Response. We agree that empirical performance and theorem scope must be separated. The updated Section 2.3 explicitly states nonnegativity, finite mean, finite positive variance, and almost-sure pairwise distinctness in Assumption 1. The final coverage theorem, Theorem 9, still invokes that assumption; no extension to arbitrary infinite-variance or tied residuals has been established in this revision.

The integrated heavy-tail experiment in Appendix C.5 identifies $t_{1.5}$ and t_2 as outside that assumption, with t_3 as the finite-variance comparison. Historical heavy-tail results remain in Appendix D.6, with their interpretation likewise qualified. At $n = 30$, TSCP's mean normalized volumes are about 1.59×10^{20} for $t_{1.5}$ and 5.92×10^{13} for t_3: near-nominal empirical coverage does not imply robust efficiency.

Signed residuals are studied through capped CQR in the body and constant shifts in the supplement. The experimental text acknowledges that capping creates ties and does not satisfy pairwise distinctness verbatim.

Author completion required. Section 2.3 proposes coordinate jitter or randomized tie-breaking, but the saved implementation perturbs scalar upper-bound scores, not coordinate residuals. Reconcile the actual implementation, the theorem's distinctness condition, and zero empirical-scale cases, or retain an explicitly empirical scope for the capped study. Qualify broad robustness language consistently.

R1.C3. Quadratic worst-case complexity and wall-clock evidence — Addressed

While the complexity discussion is appreciated, I am not fully convinced by the computational story yet. The worst-case complexity is still quadratic in the calibration size up to dimension factors, and it would help to see more direct runtime comparisons in the main paper, not only prediction-set volume comparisons. Since one of the selling points is tractability, clearer wall-clock evidence would strengthen the paper.

Response. Figure 3 now includes construction time as calibration size varies, and Figure 8 compares wall-clock times across all six retained real datasets. TSCP's mean time ranges from approximately 0.74 to 7.46 ms for calibration sizes 15–490 and dimensions 2–16. It is slower than Unscaled Max and Point CHR but faster than the empirical-copula implementation on the two 16-dimensional SCM datasets, where that baseline requires about 95–118 ms.

The timings exclude the shared regression fit and residual computation. The original comparison retains the historical measurements. A separate new diagnostic cohort in Section 4.5 adds serial, uninstrumented construction timings and directly records the search branches, answering R1.Q2 below. The theoretical range remains $O(d^2 n \log n)$ to $O(d^2 n^2)$, with thresholds reusable across test inputs. These measurements demonstrate observed cost at the tested sizes, not improved worst-case complexity.

R1.C4. Contamination and robust standardization — Addressed

The paper's robustness to outliers seems mixed. The authors acknowledge this in the discussion and point to rf2 as an example where influential observations can substantially enlarge the sets. That honesty is appreciated, but it also suggests that the current standardization scheme may be brittle in some realistic settings. I would have liked either a stronger empirical stress test on contamination/outliers or a small robust variant, even if only heuristic.

Response. We added the stronger stress test in Appendix C.6, shown in Figure S6. With $d = 10$, $n = 100$, and 200 repetitions, each observation's Gaussian noise vector is multiplied by 10 with probability $\varepsilon \in \{0, 0.01, 0.05, 0.10\}$. The same mixture generates training, calibration, and test observations. This tests exchangeable contamination, not adversarial calibration-only corruption or distribution shift.

Coverage changes from approximately 0.904 without contamination to 0.902 at 10% contamination, but TSCP's median normalized volume increases from 5.38×10^{10} to 1.10×10^{17}. Median volume is explicitly identified as the graphical summary. The study confirms efficiency sensitivity despite stable coverage, and the unfavorable `rf2` result remains visible.

We do not claim an implemented or proved robust variant. Replacing mean/standard-deviation scaling would require reconsidering link functions and worst-case calculations; it remains future work, consistent with the retained discussion.

Additional Questions

R1.Q1. Formal assumptions versus infinite-variance experiments Addressed

Can the authors clarify more explicitly the gap between the formal assumptions used in the validity results and the heavy-tailed experiments where variance may be infinite?

Response. The integrated experiments make the distinction explicit. Assumption 1 imposes finite variance and is still invoked by Theorem 9. Thus $t_{1.5}$ and t_2 in Appendix C.5 are empirical stress tests outside that theorem, not extensions of it. The same qualification applies to the historical results in Appendix D.6.

Although a generic exchangeability-based rank argument does not itself require population moments, this alone does not remove assumptions from the full TSCP construction. No moment-relaxation theorem is inferred from the simulations.

R1.Q2. Frequency and cost of backward search Addressed

How often does the algorithm fall into the less favorable backward-search regime in practice, and what are the actual runtime implications on the datasets considered?

Response. We have now measured the actual branches in 200 new reruns of each of the six datasets. The study uses the original 75%/5%/20% allocation, model hyperparameters, and hash-seeded split convention. It is explicitly identified as a new diagnostic cohort because the original fitted models were not saved. All 1,200 returned TSCP rectangles were checked against the pre-instrumentation implementation and match exactly.

At $\alpha = 0.1$, backward search occurs in 0/200 splits for every dataset; all 10,200 coordinate searches use the binary branch. Table 3 reports the split and coordinate denominators separately. Figure 10 shows approximately 4.00–8.49 inspected candidate cells per coordinate and mean whole-call construction times of 0.466–7.239 ms across datasets. Each split is timed after warm-up using seven uninstrumented calls; its median is used in the average. Fitting, residual computation, and instrumentation are excluded, and timings are collected serially after all model fits finish.

The unobserved backward regime has no estimable conditional runtime at the main target, so its entry is a dash, not zero. Zero events in 200 splits give a pointwise one-sided 95% upper bound of about 1.49% on a dataset's split-event probability; this is not a simultaneous guarantee or an analysis treating coordinates as independent.

To exercise the exceptional branches, Appendix C.10 reuses the same residuals at $\alpha = 0.3, 0.5, 0.7, 0.9$. On `energy` at $\alpha = 0.5$, 3 splits use backward search, 140 fall back, and 57 use only binary search. Mean whole-call times conditional on these regimes are 0.295, 0.121, and 0.268 ms. At $\alpha = 0.7$ and 0.9, backward search occurs in 10 of 200 energy splits. Each observed backward coordinate visits only one candidate, so the study does not demonstrate a long or worst-case backward scan. The changed targets are explicitly labeled as computational stress tests, not additional 90%-coverage benchmarks.

R1.Q3. Gains with quantile-regression residuals Addressed

Do the authors expect the gains of TSCP to persist when using quantile-regression residuals rather than absolute mean-regression residuals?

Response. The new study in Section 4.3 provides empirical evidence. All methods share fitted quantile regressions in each repetition. TSCP uses $E_j = \max\{S_j, 0\}$; Unscaled Max and Empirical Copula use signed scores; CQHR is native. The setting has three Gaussian outcomes, scales $(3, 2, 1)$, base-interval miscoverage 0.5, final target 0.9, and 30 fresh-data repetitions.

At $n = 100$, TSCP has mean coverage 0.9045 and full outcome-space volume about 2.91×10^4, compared with 0.8973 and 3.43×10^4 for Unscaled Max. Their mean full lengths are $(36.2, 36.9, 20.9)$ and $(33.3, 33.8, 29.0)$ (Figure 7). The gain is width allocation, not shortening every interval.

Across displayed sizes, TSCP coverage ranges from about 0.899 to 0.922. At $n = 50$, the mean 0.8991 has repetition-level SD 0.0457 and is retained without adjustment. Base-width and shift sensitivities in Appendix C.7 make clear that the observed gains are design-dependent, not universal.

Response to Reviewer 2

Major Comments

R2.M1. Point CHR versus CQHR and additional data splitting Open

In section 1.2 you discuss that Sampson and Chan (2024) require an additional independent dataset. This is true for their Point CHR method, but not for their quantile regression approach (CQHR).

Response. The reviewer is correct. Point CHR uses an auxiliary scale-estimation/calibration split, whereas CQHR obtains relative adjustments from fitted quantile widths without that additional split. CQHR still separates training from ordinary conformal calibration.

The integrated experiment descriptions distinguish the procedures, but Section 1.2 still attributes the additional-dataset requirement to Sampson and Chan without restricting it to Point CHR, and groups the reference with flexible nonrectangular methods.

Author completion required. Correct Section 1.2 and qualify the data-efficiency claim. Also check the sentence attributing a demonstrated comparison to Baheri and Amiri Shahbazi against the methods actually evaluated. Non-experimental wording has deliberately been left unchanged.

R2.M2. Dimension-specific comparisons Addressed

After Theorem 1 you discuss that the overall coverage cannot be used to determine if the dimension-specific coverage is tight. However, both the real data analyses and simulation studies lack dimension-specific comparisons (e.g., comparing prediction interval lengths or listing marginal coverages across the 10 dimensions in the first simulation study for TSCP). Adding these to your numerical comparisons would improve them greatly.

Response. Figure 4 now reports coordinate-wise mean full lengths in the ten-dimensional heterogeneous Gaussian setting with $\rho = 0.6$ and $n = 100$. TSCP's lengths decrease from 49.4 to 5.0 across noise scales, whereas Unscaled Max assigns approximately 38.0 to every coordinate. Joint coverage is about 0.907 and 0.902, respectively; the smaller copula bars are interpreted alongside its 0.873 joint coverage.

Figure 7 provides the quantile comparison: lengths are $(36.2, 36.9, 20.9)$ for TSCP, $(33.3, 33.8, 29.0)$ for Unscaled Max, and $(46.2, 120.5, 66.9)$ for CQHR. The partial-heteroskedasticity figure also identifies affected coordinates. All report outcome-space lengths, not transformed-score thresholds.

We have also added the previously missing aggregate real-data analysis. Figure 9 covers all 51 output coordinates across the six real datasets, using 200 paired reruns and the same fitted model and test set for every method in each split. To compare different outcome units, bars report mean within-split ratios of full lengths to Unscaled Max. Raw full lengths, their standard deviations, and target names are retained in the supporting CSV files. Infinite Point CHR intervals on `stock` are explicitly marked rather than truncated or excluded from the averages.

On `energy`, TSCP's mean full lengths are $(2.72, 9.85)$ versus $(7.77, 7.77)$ for Unscaled Max, with joint coverages 0.923 and 0.917. This is reallocation, not uniform shortening. On `student`, all TSCP length ratios exceed one, $(1.01, 1.08, 1.18)$, so the unfavorable comparison remains visible.

Figure S10 and Table S4 report the matching coordinate-wise and joint coverages. For example, TSCP's stock marginals range from 0.972 to 0.999 while joint coverage is 0.956. The supporting data contain all 40,800 split–method–coordinate records, across-split SDs, and pointwise Monte Carlo intervals. These quantify randomized-split variation conditional on each fixed dataset, not uncertainty over new populations or simultaneous coverage across all reported metrics. The original two-point illustration in Table 2 is preserved, but is no longer the sole real-data coordinate evidence.

R2.M3. Frequency of failure of the regularity condition — Addressed

In section 3.4.4 it is stated: "Under the regularity condition $\mathcal{Y}_{\boldsymbol{h}^*} \neq \varnothing$, which can be immediately verified and typically holds in practice..." How often did this not hold in the real data experiments, if at all?

Response. The new study directly instruments the mean-cell/global-rectangle intersection check that triggers fallback. At the main miscoverage level $\alpha = 0.1$, the count is 0/200 on each of `stock`, `rf2`, `scm1d`, `scm20d`, `energy`, and `student`. The measured counts are in Table 3, with protocol and uncertainty qualifications in Section 4.5.

The unit is one calibration split, not each test observation. We record the actual branch; equality between local and global output rectangles is not used as a proxy. Zero observed events do not imply impossibility: the pointwise one-sided 95% upper bound on a fixed dataset's split-event probability is approximately 1.49%. This does not pool dependent coordinates or provide a simultaneous bound across datasets.

The computational stress sweep in Appendix C.10 and Figure S11 shows that fallback is observable at more extreme targets. At $\alpha = 0.9$, the counts are 139, 185, 180, 186, 190, and 200 out of 200, respectively, in the dataset order above. This supports the restricted empirical observation at the main target, not a universal claim that the condition always holds. Fallback remains distinct from backward search throughout the tables and saved trial records.

R2.M4. Missing CQHR baseline — Addressed

The numerical studies do not include a comparison with CQHR from Sampson and Chan (2024), which performed better than Point CHR in their paper.

Response. Native CQHR is now a main-text comparator in Section 4.3 and Figure 6. Every method shares fitted lower/upper gradient-boosted quantile models, trained on 2,400 observations and evaluated on 600 fresh test observations. The models use 100 stages, depth 5, learning rate 0.05, and minimum leaf size 5. We use 30 repetitions and calibration sizes $30, 50, 100, 200$.

Endpoints are ordered pointwise. CQHR uses signed scores, the first coordinate as reference, and a 10^{-12} floor on widths in scale ratios; these choices are disclosed. At $n = 100$, its mean coverage is 0.8936 (SD 0.0318) and mean volume 1.29×10^8 (SD 3.79×10^8), versus coverage 0.9045 (SD 0.0374) and volume 2.91×10^4 for TSCP.

Small empirical coverage differences are not treated as violations of a theoretical guarantee. The highly variable CQHR volumes are specific to the fitted models and width ratios, not evidence of universal inferiority. The coordinate plot and base-miscoverage sweep in Figure S7 provide additional context.

R2.M5. Missing convex shape-template baseline and related work — Partially addressed

The numerical studies and lit review also fail to compare TSCP with any of the approaches from Multi-Modal Conformal Prediction Regions with Simple Structures by Optimizing Convex Shape Templates (Tumu et al. 2024), which can be used to form convex (including hyper-rectangular) prediction regions.

Response. The requested family is now compared in Appendix C.8 and Figure S9. The implementation uses public conformal-region-designer components: KDE with 20 grid locations per dimension, mean-shift clustering, and rectangular templates. Calibration residuals are split equally between template learning and conformalization. Score-consistent inflation and a 0.05 aspect-ratio floor are disclosed; this is not an undisclosed unmodified-package comparison.

The two-dimensional Gaussian setting uses scales $(2, 1)$, calibration sizes $30, 50, 100, 200$, and 100 repetitions. At $n = 100$, template coverage is about 0.904, normalized volume 10.07, and construction time 98.7 ms, compared with 0.906, 8.04, and 0.80 ms for TSCP. Restriction to low dimension reflects this implementation's 20^d Cartesian grid, not a limitation of every template approach. A unimodal Gaussian study does not fully test the advantages of multimodal regions.

Author completion required. The citation is present in the new experiment and bibliography, but Section 1.2 still lacks the related-work discussion. Add it and qualify the suggestion that competing learned geometries are necessarily nonrectangular.

R2.M6. Heteroskedasticity in only some coordinates — Addressed

Standardizing the residuals to be used as scores is done to address heterogeneity across dimensions. How does it work if some, but not all, dimensions have heteroskedastic errors? A simulation demonstrating this would make the numerical experiments more interesting.

Response. We added the main-text experiment in Figure 5, where only the first five of ten coordinates are heteroskedastic:

$$\sigma_j(x_1) = (11 - j)\sqrt{\frac{1 + 1.5x_1^2}{2.5}}\ (j \le 5), \qquad \sigma_j(x_1) = 11 - j\ (j > 5).$$

For $X_1 \sim \mathcal{N}(0, 1)$, this preserves $\mathbb{E}[\sigma_j^2(X_1)] = (11 - j)^2$, separating input-dependent variability from a change in marginal noise variance.

At $n = 100$, mean coverage and normalized volume are approximately 0.904 and 1.40×10^{11} for TSCP, 0.902 and 2.94×10^{11} for Point CHR, and 0.900 and 2.33×10^{13} for Unscaled Max. The coordinate profile identifies the heteroskedastic outputs. The protocol matches the other new absolute-residual studies: 200 repetitions, 7,200 training observations, and 800 test observations.

The text explicitly limits interpretation to joint marginal coverage. Global scaling does not estimate the local variance function or establish coverage conditional on every X_1.

R2.M7. Absolute-value CQR scores, constant shifts, and choosing the constant — Partially addressed

Above equation 9, you state one possible score is the absolute value of CQR. It seems like this score would lead to a larger adjustment than the standard CQR score, which cannot be used because of Assumption 1. Is that not the case? You mentioned earlier in the paper that a constant could be added to the standard CQR scores so they are non-negative, and not overly conservative when the quantile regression coverage is conservative. Would it not be beneficial to include this score instead of the absolute value of the CQR score, so that practitioners don't use an overly conservative approach? Along the same line, it would be nice to know how to choose such a constant in practice, so that this approach could be used with quantile regression.

Response. The experiments now distinguish the transformations. For signed CQR score

$$S_j(x, \boldsymbol{y}) = \max\{\widehat{q}_{\ell,j}(x) - y_j, y_j - \widehat{q}_{u,j}(x)\},$$

the body uses $E_j = \max\{S_j, 0\}$, not $|S_j|$. At a nonnegative threshold W_j, it inverts to $[\widehat{q}_{\ell,j}(x) - W_j, \widehat{q}_{u,j}(x) + W_j]$. Capping avoids treating an observation deep inside the base interval as a large positive error, but cannot shrink a conservative base interval. The inequality $\max\{S_j, 0\} \le |S_j|$ is not asserted to order all adaptively standardized final volumes. The explicit absolute-value CQR example from the old Section 2.5 has been removed in the supplied general set-valued formulation.

Shifted-score experiment. Appendix C.7 and Figure S8 use $S_j + C$, apply TSCP, and subtract C from its coordinate adjustment. With $C = 100, 300, 1000$, $d = 3$, $n = 100$, and 30 repetitions, coverage is approximately 0.9077 and mean volume 2.95×10^4. Within each trial and tested constant, recorded coordinate lengths equal shifted TSCP-GWC's. Across constants, the largest paired discrepancy is about 2.2×10^{-12}. This is a numerical finding for the tested design, not a universal invariance theorem.

Selecting a shift. For ordered base widths $w_j(x)$, $S_j(x, \boldsymbol{y}) \geq -w_j(x)/2$. A known uniform bound $w_j(x) \leq M$ permits $C \geq M/2$. It must be valid on the relevant input domain and independent of calibration/test responses, for example justified from the trained model and a domain bound. An observed calibration minimum alone is insufficient for future inputs. The fixed-constant sweep is not a validated general selector.

Author completion required. Section 2.3 still describes the shift as preserving the resulting prediction sets without qualifying the practical local approximation. Its tie-breaking discussion also does not resolve the saved implementation's coordinate ties and zero-scale cases. Reconcile these claims with the actual construction before treating this point as fully addressed. The experimental discussion already states the limitations.

Minor Comments

R2.m1. Marginal coverage labels in Figure 1 — Open

In Figure 1 it would be a nice addition to include the marginal coverage rates for the rightmost image. Making the marginal coverages be different (e.g., 92% and 93%) would also highlight early on that the marginal dimension coverages need not be the same.

Response. Joint coverage need not equal either coordinate's marginal coverage, and the two marginals need not match. However, Figure 1 remains the original schematic: its rightmost panel labels joint coverage only.

Author completion required. Add both coordinate marginal labels to the figure and drawing source. If values such as 92% and 93% are chosen illustratively, identify them as schematic rather than computed from the unchanged plot. This non-experimental figure has not been edited.

R2.m2. Meaning of "uniformly tight" in the population oracle — Open

In section 2.5 when discussing the oracle prediction set your second claim is: "it tends to be uniformly tight across all output dimensions, because it operates with residuals whose components are all on the same scale". Does tight here refer to the marginal coverage, or something else? If it is the coverage, would the error distribution shapes need to be the same for this to be true?

Response. Matching means and variances does not imply equal marginal coverage or equal tail shapes. The intended benefit is coordinate-adaptive width, not a distribution-free balance or minimum-volume guarantee. Identical standardized marginal distributions give identical coverage at a fixed common cutoff; a random data-dependent cutoff needs additional symmetry or an appropriate asymptotic argument.

Section 2.5 still uses *uniformly tight* with the same rationale, and related language remains in Section 2.4.

Author completion required. Define or replace the term: specify whether it means width, volume, oracle approximation, or marginal coverage. Retain a balance claim only with the requisite assumptions and justification. The new bar plots illustrate width allocation, not equal marginal coverage.

R2.m3. Labeling the boundaries in Figure 2 Open

In section 3.4.4, after you introduce L_j, you discuss what L_1 and L_2 are with respect to Figure 2. Adding L_1 and L_2 to the margins (or with a dashed line across the plot) would be a helpful visualization.

Response. Figure 2 still lacks endpoint labels. The revised method denotes the residual boundaries by $\mathcal{L}_1, \mathcal{L}_2$, and the figure should use that convention rather than the old L_j notation.

Author completion required. Add margin labels or guide lines for $\mathcal{L}_1, \mathcal{L}_2$ to the schematic and its drawing source, identifying residual-space axes. For absolute scores, full outcome lengths are $2\mathcal{L}_j$. This protected methodology figure has not been changed.

R2.m4. Runtime comparisons in the main text Addressed

The authors discuss the computational cost of using TSCP as they introduce it. There are some plots in the appendices which compare the computational cost. They should either add a note in the numerical studies that computational cost comparisons can be found in the appendix, or (better) include similar plots in the main text.

Response. We followed the suggestion to place runtime evidence in the body. Figure 3 includes construction time, and Figure 8 compares all six real datasets in milliseconds on a logarithmic axis. Additional timings are in Appendix C.2 and Appendix C.8.

The old approximation and dimension-scaling runtime studies are retained in Appendix D.5; no prior runtime figure was deleted. Captions specify that common model fitting and residual computation are excluded. The text neither claims that TSCP beats every simpler baseline nor that empirical timings improve its worst-case complexity.

R2.m5. Infinite Point CHR volume on the stock dataset Addressed

In the stock data analysis, the volume of Point CHR is infinite because of the size of the training/calibration sets along with the coverage rate. One would not split their data in such a way in practice, which should be stated. The comparison does make the point that Point CHR does not use the data as efficiently, but it should be clarified that the infinite prediction sets are caused by how small the last calibration set size is, and not the method.

Response. The main real-data discussion now attributes the infinite Point CHR volume on `stock` to the chosen allocation, not an intrinsic failure of the method. The common 75%/5%/20% split leaves only 15 calibration observations before the additional Point CHR split. Its final subset is too small for a finite 0.9 conformal quantile.

With m final-stage observations, the rank is $\lceil (1-\alpha)(m+1) \rceil$; if it exceeds m, the augmented $+\infty$ is selected. At $\alpha = 0.1$, a finite rank needs at least $m = 9$. A practitioner would allocate more data to that stage, at a cost elsewhere.

Appendix C.4 and Figure S4 separately report the fraction of infinite regions: Point CHR is infinite throughout the displayed $n = 10$ experiment and finite throughout the displayed $n \geq 20$ experiments. This remains a common-allocation comparison, not a separately retuned Point CHR benchmark. Both the original real-data table and its new presentation are retained.

R2.m6. Vector and scalar notation Partially addressed

Readability would be improved if vectors were bolded while scalars were not.

Response. The supplied revision substantially improves vector/scalar typography in the setup, residual definitions, and parameter vectors. It also distinguishes outcome endpoints L_j, U_j from residual thresholds $\mathcal{L}_j$, which the integrated experiments follow.

The convention remains inconsistent: the mean-index vector h^* and related multi-indices are unbolded in Section 3.4.4 and algorithms, while coordinate quantities $\boldsymbol{\mu}_j^*$ and $\boldsymbol{\sigma}_j^*$ remain bold despite being scalars. Some algorithmic vector means/scales are also unbolded.

Author completion required. Complete the notation pass through Sections 2–3, algorithms, captions, and proofs. The actual source supports a partial improvement, not a completed global correction.